Program overview

19-May-2025 11:44

Year	**2024/2025**
Organization	**Civil Engineering and Geosciences**
Education	**Schakelopleiding Transport, Infrastructure and Logistics**

Code	Omschrijving	ECTS	p1	p2	p3	p4	p5
Schakelopleiding TIL 2024	**Bridging Program TIL 2024**						
Schakelopleiding TIL 2024	**Bridging Program TIL 2024**						
CTB1420-17	Transport en Planning	5					
CTB1420-17 Toets 1	*Deeltentamen Delen 1 en 2 van het dictaat*	*0*					
CTB1420-17 Toets 2	*Deeltentamen Delen 3 en 4 van het dictaat*	*0*					
CTB1420-17 Toets 3	*Ethics CTB1420-17*	*0*					
CTB1420-17 Toets 4	*Oefening CTB1420-17*	*1*					
CTB1420-17 Toets 5	*Integrale Toets Delen 1 t/m 5*	*4*					
IFEEMCS010400	Linear Algebra	5					
IFEEMCS010500	Probability and Statistics	3					
IFEEMCS012100	Calculus for Engineering, part 1	3					
IFEEMCS012200	Calculus for Engineering, part 2	3					
IFEEMCS012300	Calculus for Engineering, part 3	3					
TB111C	Problem Analysis	5					
WI1909TH	Differential Equations	3					

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Schakelopleiding TIL 2024

Year	**2024/2025**
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Schakelopleiding TIL 2024	
Contact for students	Dr.ir. J.H. Baggen (j.h.baggen@tudelft.nl)
Director of Education	Dr.ir. D.L. Schott
Program Coordinator	Dr.ir. J.H. Baggen (j.h.baggen@tudelft.nl)
Introduction 1	Students with a relevant Dutch higher vocational institute bachelors degree have to complete a bridging pre-master programme before they will be admitted to the masters degree programme. The bridging programme as referred to in paragraph 1, comprises the following deficiency subjects amounting to 31 credits in total.

CTB1420-17	**Transport en Planning**	**5**
Responsible Instructor	Dr. V.L. Knoop	
Instructor	Dr.ir. N. van Oort	
Practical Coordinator	Dr.ir. A. Gavriilidou	
Education Period	4	
Start Education	4	
Exam Period	4	
Course Language	Dutch	
Collegerama	No	

CTB1420-17 Toets 1	***Deeltentamen Delen 1 en 2 van het dictaat***	***0***
Responsible Instructor	Dr. V.L. Knoop	
Education Period	4	
Start Education	4	
Exam Period	4	
Course Language	Dutch	
Collegerama	No	

CTB1420-17 Toets 2	***Deeltentamen Delen 3 en 4 van het dictaat***	***0***
Responsible Instructor	Dr. V.L. Knoop	
Education Period	4	
Start Education	4	
Exam Period	4	
Course Language	Dutch	
Collegerama	No	

CTB1420-17 Toets 3	***Ethics CTB1420-17***	***0***
Responsible Instructor	Dr. V.L. Knoop	
Education Period	4	
Start Education	4	
Exam Period	4	
Course Language	Dutch	
Collegerama	No	

CTB1420-17 Toets 4	***Oefening CTB1420-17***	***1***
Responsible Instructor	Dr. V.L. Knoop	
Education Period	4	
Start Education	4	
Exam Period	4	
Course Language	Dutch	
Collegerama	No	

CTB1420-17 Toets 5	***Integrale Toets Delen 1 t/m 5***	***4***
Responsible Instructor	Dr. V.L. Knoop	
Education Period	4	
Start Education	4	
Exam Period	4	
Course Language	Dutch	
Collegerama	No	

IFEEMCS010400	Linear Algebra	5
Responsible Instructor	Dr. E. Emsiz	
Contact Hours / Week x/x/x/x	6/x/x/x	
Education Period	1	
Start Education	1	
Exam Period	1 2	
Course Language	Dutch English	
Expected prior knowledge	See the requirements of the specific MSc programma students are applying for.	
Summary	Course email: ifeemcs010400@tudelft.nl (please don't use personal email addresses)	
Course Contents	- Solving systems of linear equations (using Gaussian elimination) - Vectors in R^n - Linear dependence/independence - Matrix algebra (sum, product, inverse) - LU decomposition of a matrix - Linear transformations - Subspaces in R^n (basis, dimension); column space and null space - Determinants - Eigenvalues and eigenvectors - Diagonalization - Inner products and orthogonal sets - Orthogonal projections - Method of least squares and linear models - Symmetric matrices	
Study Goals	An important goal of the course is to make the student comfortable with fundamental notions from linear algebra. At the end of this course you will be able to understand the connections between these concepts. You may also be asked to come up with proofs for elementary properties (i.e. write down a consistent/correct short proof) and give a correct counterexample. After successfully completing the course you will be able to: - Solve systems of linear equations. - Identify if a system of linear equations is solvable, and if the solution is unique. - Show the equivalence between a vector equation, a matrix equation and a system of linear equations. - Give a geometric description of solutions sets of vector equations. - Determine the linear (in)dependence of a set of vectors and identify the (geometric) properties of linear dependence. - Identify if a transformation is linear and determine the standard matrix of a linear transformation. - Determine the matrix of geometric linear transformations. - Be able to perform matrix operations (sum, scalar multiple, multiplication, transpose) and describe the law-like properties of matrix multiplication and apply these properties. - Describe the properties of invertible matrices, determine the inverse of a non-singular matrix and apply these topics. - Give definitions of subspace, column space and null space of a matrix and determine a basis for a subspace. - Describe and apply the concept of dimension of a linear subspace, rank of a matrix and the rank theorem. - Determine the LU-decomposition of a matrix. - Determine the determinant of a matrix by applying the properties of a determinant. - Apply the properties of determinants (product, transpose, inverse). - Determine the eigenvalues and eigenvectors of a matrix. - Determine the algebraic and geometric multiplicity of a matrix. - Determine if a matrix is diagonalizable and if so, determine a diagonalization of the matrix. - Analyze the case of complex eigenvalues and eigenvectors. - Determine the matrix of a linear transformation with respect to a basis. - Calculate and apply properties of the inner product in the context of orthogonality and orthogonal sets. - Calculate the orthogonal projection on a subspace. - Determine an orthogonal basis of a subspace by means of the Gram-Schmidt process. - Explain the normal equation of the least-squares method and apply the least-squares method to linear models. - Explain and calculate the orthogonal diagonalization of symmetric matrices and describe the spectral theorem for symmetric matrices.	
Education Method	Colstruction	
Literature and Study Materials	Book: David C. Lay et. al., Linear Algebra and its Applications, Global Edition, 6/E. - Slides on Brightspace. - Prelecture videos. - Online homework assignments. - Book exercises - Old exams on Brightspace. Disclaimer: information may change depending on the developments around the coronavirus.	
Assessment	Final score: F IF the midterm score M is higher than the score E for the exam AND E >= 5.0 THEN F = 0.25 M + 0.75 E ELSE F = E All grades are rounded to 1 decimal place. Note that the midterm is not mandatory (and cannot be resited). For the resit the midterm is not relevant anymore.	
Permitted Materials during Tests	None	
Test and result scale	1-10	
Co-Instructor	Dr.ir. R.F. Swarttouw	

IFEEMCS010500	Probability and Statistics	3
Responsible Instructor	Dr. C. Kraaikamp	
Responsible Instructor	Dr. R.J. Fokkink	
Contact Hours / Week x/x/x/x	x/x/4/x or x/x/x/4	
Education Period	3 4	
Start Education	3 4	
Exam Period	3 4 5	
Course Language	Dutch English	
Expected prior knowledge	Linear Algebra, Calculus for Engineering part 1 and part 2 or Calculus for Science part 1 and part 2.	
Course Contents	Graphical and numerical summaries of data Introduction to probability Some elementary combinatorics to find probabilities Conditional probability; Bayes' rule; stochastic (in)dependence Random variables; probability mass function; density function; distribution function Standard distributions: Binomial, Poisson, Geometric, Normal, Uniform, Exponential, Pareto Expectation and variance; transformations; Jensen's inequality Multivariate random variables; joint distributions; joint and marginal density functions; (in)dependence of random variables Covariance and correlation Chebychev's inequality; Law of Large numbers; Central Limit Theorem Sampling theory and statistical models; mean, sample variance, histogram, empirical distribution function, boxplot Theory of Estimators: Bias, Efficiency, Mean squared error Linear Regression: univariate and multivariate, goodness of fit Introduction to statistical learning Confidence intervals of Mean and Proportion; t-distribution Testing theory: Type I/II Error, p-value, significance level, critical region One sample t-test	
Study Goals	At the end of the course, you will be able to: - compute (conditional) probabilities - use discrete and continuous random variables - use joint random variables - compute expectation and (co)variance - apply the law of large numbers and the central limit theorem - analyse data by means of graphical and numerical summaries - formulate a statistical model - estimate parameters - perform a linear regression analysis - construct a confidence interval - formulate and carry out hypothesis tests - Use a computer program to perform elementary simulation and data analysis	
Education Method	Interactive lectures, pre-lecture videos, exercises	
Literature and Study Materials	Book: Mathematical introduction to Probability and Statistics, Dekking et al.	
Assessment	One midterm (week 5) on Part 1: Probability Theory and one endterm (week 10) on Part 2: Statistics. Final grade is the average of both tests. To pass the course, the grade for part 2 needs to be at least 4 or higher.	
Permitted Materials during Tests	1) Formula sheet to be used for exams of Probability and Statistics 2) Any kind of calculator	
Remarks	Please note that this course is available in Q3 and in Q4.	
Co-Instructor	Dr. C. Kraaikamp	
Co-Instructor	Dr. R.J. Fokkink	

IFEEMCS012100	Calculus for Engineering, part 1	3
Responsible Instructor	Dr. E. Emsiz	
Contact Hours / Week x/x/x/x	4/x/x/x	
Education Period	1	
Start Education	1	
Exam Period	1 2	
Course Language	Dutch English	
Expected prior knowledge	See the requirements of the specific MSc programma students are applying for.	
Co-Instructor	Dr.ir. N.P.K. Chan	

IFEEMCS012200	Calculus for Engineering, part 2	3
Responsible Instructor	Dr. E. Emsiz	
Contact Hours / Week x/x/x/x	x/4/x/x	
Education Period	2	
Start Education	2	
Exam Period	2 3	
Course Language	Dutch English	
Co-Instructor	Dr.ir. N.P.K. Chan	

IFEEMCS012300	Calculus for Engineering, part 3	3
Responsible Instructor	Dr.ir. R.F. Swarttouw	
Contact Hours / Week x/x/x/x	x/x/4/x	
Education Period	3	
Start Education	3	
Exam Period	3 4	
Course Language	Dutch English	
Expected prior knowledge	Calculus for Engineering 2 (IFEEMCS012200) or a similar course.	
Course Contents	Line integrals and surface integrals. Integral theorems of Green, Stokes and Gauss. Sequences and series. Power series and Taylor series.	
Study Goals	After this course the student should be able to 1. find parametrizations of curves and surfaces (in 2D and 3D); 2. calculate a line integral or a surface integral of a function or a vector field; 3. mention the physical interpretation of these line and surface integrals; 4. define and formulate the physical meaning of concepts like gradient, curl and divergence; 5. calculate line and surface integrals via one of the integral theorems (Green, Stokes, Gauss) 6. determine whether a sequence is convergent or divergent, for example by calculating its limit; 7. determine whether a series is convergent or divergent with a test for convergence: Integral Test, (Limit) Comparison Test, Ratio or Root Test; 8. determine the interval of convergence of a power series; 9. determine the Taylor series of a simple analytic function, for example sin(x), cos(x), exp(x), ln(1+x), arctan(x), (1+x)^a.	
Education Method	Lectures (per week 2x2 hours).	
Books	Calculus, Volume 1, 2 and 3. Edwin Herman, Gilbert Strang. ISBN-13 978-1-947172-13-5, 978-1-947172-14-2, 978-1-947172-16-6 This is an open source book that also will be made available (for free) via Brightspace.	
Assessment	On-campus exams. Partly digital, partly written. See Brightspace for more details. There will be two partial exams, of 90 minutes each: - The first partial exam is about vector calculus; - The second partial exam is about sequences and series. The final grade is the average of the grades for the two partial exams. There is only one resit, about the complete course, that lasts 180 minutes. It is NOT possible to resit only one partial exam.	
Permitted Materials during Tests	A formula sheet (see Brightspace). NO calculators allowed.	
Co-Instructor	Dr. F.J. van de Bult	

TB111C	Problem Analysis	5
Module Manager	N. van der Linden	
Instructor	M. Looij	
Instructor	N. van der Linden	
Responsible for assignments	N. van der Linden	
Co-responsible for assignments	Dr.ir. M.P.M. Ruijgh-van der Ploeg	
Contact Hours / Week x/x/x/x	5/0/0/0	
Education Period	1	
Start Education	1	
Exam Period	1	
Course Language	Dutch	

WI1909TH	Differential Equations	3
Responsible Instructor	Drs. I.A.M. Goddijn	
Contact Hours / Week x/x/x/x	0/4/0/0	
Education Period	2	
Start Education	2	
Exam Period	2 3	
Course Language	Dutch	
Expected prior knowledge	Calculus and Linear Algebra	
Course Contents	Linear differential Equations, Systems of first order, linear and not-linear differential equations, Partial differential equations and Fourier series.	
Study Goals	Learning a number of mathematical techniques necessary for solving various technical-physical model equations.	
Education Method	Colstruction	
Books	Elementairy Differential Equations and Boundary Value Problems, Global Edition ISBN-13: 978-1-119-38287-4 (preferable) or Elementairy Differential Equations and Boundary Value Problems, International Adaptation ISBN-13: 978-1-119-82051-2	
Assessment	Exam with open questions. Due to circumstances (such as Covid-19) this can become an online exam, even in a different format. In case of insufficient results a repair option may exist in accordance with Article 2, Examination requirements, Clause 4, of the Implementation Regulations 2023-2024.	
Permitted Materials during Tests	Announced during lectures and on Brightspace	
Co-Instructor	Dr.ir. Z.J.G. Gromotka	

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